



DATA SCIENCE ROADMAP

Step by Step Guide to Become a Data Scientist

LEARN → PRACTICE → BUILD → GROW



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1. Programming Basics

- Learn Python Programming
- Core Concepts: Variables, Loops, Functions
- Data Structures (Lists, Tuples, Dicts, Sets)
- Object Oriented Programming (OOP)

2-4 Weeks

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2. Mathematics & Statistics

- Mathematics: Algebra, Calculus, Linear Algebra
- Statistics: Descriptive & Inferential Statistics
- Probability Distributions & Hypothesis Testing
- Correlation, Regression Basics

4-6 Weeks

3



3. Data Handling & SQL

- Data Types & Structures
- Data Cleaning & Manipulation (Pandas Basics)
- SQL Basics: SELECT, WHERE, JOIN, GROUP BY
- Aggregations, Subqueries, CTE

2-3 Weeks

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4. Data Analysis & Visualization

- Exploratory Data Analysis (EDA)
- Data Visualization: Matplotlib, Seaborn
- Business Insights from Data
- Advanced Visualization: Plotly

2-4 Weeks

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5. Machine Learning Basics

- What is Machine Learning?
- Types: Supervised, Unsupervised, Reinforcement
- Train, Test, Validation Split
- Overfitting vs Underfitting

3-4 Weeks

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6. Machine Learning Algorithms

- Linear Regression
- Logistic Regression
- Decision Trees & Random Forest
- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)

4-6 Weeks

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7. Advanced Machine Learning

- Ensemble Techniques (Bagging, Boosting, XGBoost)
- Dimensionality Reduction (PCA)
- Clustering (K-Means, Hierarchical)
- Model Evaluation & Cross Validation

3-4 Weeks

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8. Deep Learning Basics

- Neural Networks Basics
- Activation Functions, Loss Functions
- Optimization Algorithms
- Building your first Neural Network

3-4 Weeks

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9. Specialized Topics (Optional but Important)

- Natural Language Processing (NLP)
- Computer Vision (CV)
- Time Series Analysis
- Recommendation Systems

3-6 Weeks

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10. Projects & Practice

- Work on End-to-End Projects
- Participate in Kaggle Competitions
- Build a Strong Portfolio on GitHub
- Case Studies & Real-world Datasets

4-6 Weeks

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11. Deployment & MLOps Basics

- Introduction to Deployment
- Streamlit / Flask for ML Apps
- Docker Basics
- Model Monitoring & Maintenance (Basics)

2-3 Weeks

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12. Continuous Learning & Growth

- Stay Updated with Latest Trends
- Read Research Papers, Blogs
- Learn New Tools & Frameworks
- Build, Share & Grow Your Network

Ongoing

ESSENTIAL SKILLS

- Analytical Thinking
- Problem Solving
- Statistics Knowledge
- Programming
- Data Visualization
- Communication

IMPORTANT TOOLS

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib / Seaborn
- Scikit-learn
- SQL
- Tableau / Power BI

DATA SCIENCE WORKFLOW



BEST PLACES TO PRACTICE

- Kaggle
- HackerRank
- LeetCode (SQL)
- DataCamp
- Kaggle Datasets
- UCI ML Repository

CAREER PATHS

- Data Scientist
- Machine Learning Engineer
- Data Analyst
- Business Intelligence Analyst
- AI Research Engineer
- Data Engineer

LEARNING RESOURCES

- Books: Hands-On ML, ISLR, Python DS Handbook
- Courses: Coursera, edX, DataCamp, fast.ai
- YouTube Channels: StatQuest, Krish Naik, Alex The Analyst
- Blogs: Towards Data Science, Analytics Vidhya



Consistency + Practice + Curiosity = Success in Data Science